19 A triangle has vertices P, Q and R.

The coordinates of P are $(-3, -6)$
The coordinates of Q are $(1, 4)$
The coordinates of R are $(5, -2)$

M is the midpoint of PQ.
N is the midpoint of QR.

Prove that MN is parallel to PR.
You must show each stage of your working.

Midpoint: average the x coordinates and the y coordinates.

$M = ((-3 + 1) \div 2, (-6 + 4) \div 2) = (-1, -1)$

$N = ((1 + 5) \div 2, (4 - 2) \div 2) = (3, 1)$

1 mark
Find the coordinates of the midpoints

Gradient = change in y ÷ change in x.

$MN: \quad \frac{1 - (-1)}{3 - (-1)} = \frac{2}{4} = \frac{1}{2}$

1 mark
Use the gradient formula

$PR: \quad \frac{-2 - (-6)}{5 - (-3)} = \frac{4}{8} = \frac{1}{2}$

1 mark
Both gradients are ½

MN and PR have the same gradient and are distinct lines.

Therefore MN is parallel to PR.

1 mark
State the conclusion from the equal gradients

(Total for Question 19 is 4 marks)

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